José Carter Arriagada

Research Software Engineer | GPU Computing, Real-Time XR & HPC

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Research Interests

GPU and parallel computing for real-time and scientific workloads; real-time rendering and XR systems, including Gaussian splatting and neural rendering on consumer GPUs; performance engineering for simulation and physical AI; high-performance computing infrastructure for research.

Education

Pontificia Universidad Católica de Chile Mar 2020 – Jul 2025

B.S. in Computer Science Engineering; Major in Software Engineering; Minor in Data Science *Santiago, Chile*

- Graduated with Distinction. Coursework and teaching in Operating Systems, High-Performance Computing, Networks, and Extended Reality.

Research Experience

Multiplatform Full Stack Software Engineer, XR Lab Sep 2026 – Present

DICTUC - Pontificia Universidad Católica de Chile, with Prof. Leonel Merino (full time) *Santiago, Chile*

- Lead software for **ANID project 13250116**, *Enhancing Collaboration in Extended Reality*: a three-headset mixed-reality study tool for research on emotion and collaboration.
- Moved the study app to a native **VIVE Focus Vision** build (**89.5 fps**, passthrough, hand and eye tracking) and built a web panel that runs every headset from a PC or phone.
- Synchronized gaze, motion, and task events with **MindWare BioLab** physiological recordings to within a few milliseconds, with no turns lost on Wi-Fi drops, so sessions are usable for analysis.
- Built **Waves AR**, an iPhone and iPad app that teaches ocean waves on the coast (Swift, Metal FFT ocean, ARKit/RealityKit), and a wireless dev loop that cut an AR iteration from about 8 minutes to 4–7 seconds.

Research Software Engineer Jul 2023 – Jul 2025

Politiktok (Fondecyt) - Pontificia Universidad Católica de Chile *Santiago, Chile*

- Led the engineering of a GPU-backed **Python/PostgreSQL** research platform processing **100,000+ records**, with about **10x faster** pipelines through parallelization and query optimization.
- Built reproducible ingestion, cleaning, transcription, feature-engineering, NLP, and visualization workflows for social-media research, supporting a peer-reviewed Springer publication.

Simulation Engineer Jan 2023 – Jul 2024

Pontificia Universidad Católica de Chile - Geoscience *Santiago, Chile*

- Developed GPU-accelerated scientific software for tectonic-plate modeling and seismic-risk analysis, optimizing **CUDA** parallel algorithms for a **15x speedup**.
- Ran large simulations on Linux workstations and university servers, and processed and visualized the resulting data; presented the work at the SSA 2023 Annual Meeting.

AI Systems Developer, UC Bridge Program Jan 2024 – Mar 2024

Drone Response - University of Notre Dame *Notre Dame, IN, USA*

- Selected among 200+ candidates for an international research program on NASA- and NSF-funded projects.
- Implemented mission-planning logic for autonomous multi-drone coordination in emergency response: Python route optimization (Hamiltonian and clustering approaches), MQTT event-driven communication, OpenAI API, and Java Spring services.

Publications & Presentations

Journal article (2026) — Co-author. *Political Subjectification on TikTok: A Postdigital Mode of Inquiry for Tracing Affects Among Chilean Youth.* Postdigital Science and Education, Springer. doi:10.1007/s42438-026-00638-4. Built the technical pipeline for large-scale data mining, transcription, and analysis.

Conference presentation (2023) — Co-author. *Interactions Between Megathrust and Adjacent Crustal Faults.* Seismological Society of America Annual Meeting, San Juan, Puerto Rico.

Teaching

Senior Teaching Assistant & Technical Mentor Mar 2023 – Jul 2025

Department of Computer Science, Pontificia Universidad Católica de Chile *Santiago, Chile*

- 14+ teaching-assistant roles for 100+ students in **Operating Systems**, **High-Performance Computing**, Networks, Extended Reality, introductory computing, and software testing; mentored systems programming and optimization.
- Named Senior Teaching Assistant through the department's teaching-assistant career program (2025).

Programming Instructor Jul 2024 – Jul 2026

Escuela Militar de Chile *Santiago, Chile*

- Designed and taught Python and AI-application courses, assessments, and materials; recognized in 2026 for dedication and professionalism.

Industry & Applied Experience

Freelance Software Engineer, Physical AI & Robotics Sep 2026 – Present
TEK Chile (part time) *Santiago, Chile*
- Perception, control, and simulation software for a Dobot CR5S roll-packing cell and a Dobot Rover X1 inspection robot: Isaac Sim digital twin, YOLO26 vision, and Modbus/TCP control.
- Rover: 18 real patrol runs with no contact, video raised from 5 to about 20 fps, Pi latency cut from 125 to about 25 ms, and a 0.48 s emergency stop.

Full Stack Developer Sep 2025 – Sep 2026
Dily - Automotive Fintech Platform *Santiago, Chile*
- End-to-end lending software with TypeScript/Node.js, Angular/Nx, PostgreSQL/MySQL, and AWS, using DDD and hexagonal architecture.

Lead Engineer, Alerting Platform Migration 2026
GridWorks *Santiago, Chile*
- Led the staged rebuild of an industrial alerting platform (Next.js, Node.js, MQTT, PostgreSQL) without interrupting live workflows.

Software Engineer Dec 2024 – Jul 2025
Flair - Commercial-Building Energy Optimization *Santiago, Chile*
- Python microservices and InfluxDB/DynamoDB time-series workflows on AWS for a platform tied to HVAC energy reductions of up to 50%.

Conferences, Programs & Awards

SC26 Early Career Program — Selected participant with travel grant covering registration, airfare, and lodging; also matched in the SC26 Mentor-Protégé Program (Chicago, Nov 2026).

ACM SIGGRAPH 2026 Student Volunteer — Supported conference operations in Los Angeles (Jul 2026); visited Treyarch through the volunteer program.

10th South American Space Generation Workshop — Space Generation Advisory Council delegate (2026).

KHIPU 2025 — Latin American Meeting in Artificial Intelligence.

Skills & Languages

Programming: Python, C++, CUDA, C#, Swift, TypeScript/JavaScript, Java, SQL
GPU & Real-Time: CUDA, Metal, Unity, OpenXR, ARKit/RealityKit, Isaac Sim, eye and hand tracking, profiling
Data & ML: PyTorch, NLP, PostgreSQL, InfluxDB, data pipelines, scientific visualization
Systems: Linux, Docker, AWS, MQTT, Git, CI/CD
Languages: Spanish (native), English (advanced, TOEIC 905)